CEGE0024

Structures and Materials: Plastic Analysis and Finite Element Modelling

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Exam information

Centrally managed exam

Duration: 2hrs

Format: 100 Marks total. Split into:

1. Structures portion 50 Marks (~1hr)
2. Materials portion 50 Marks (~1hr)

Structures:

- Covering weeks 1-5.
- Two questions
- Similar difficulty to previous years exams
- *Having done part 1 of the assignment you should have a good foundation!*

Ninth session objectives

- Verification
- Notes on the assignment

Modelling process

BOOK

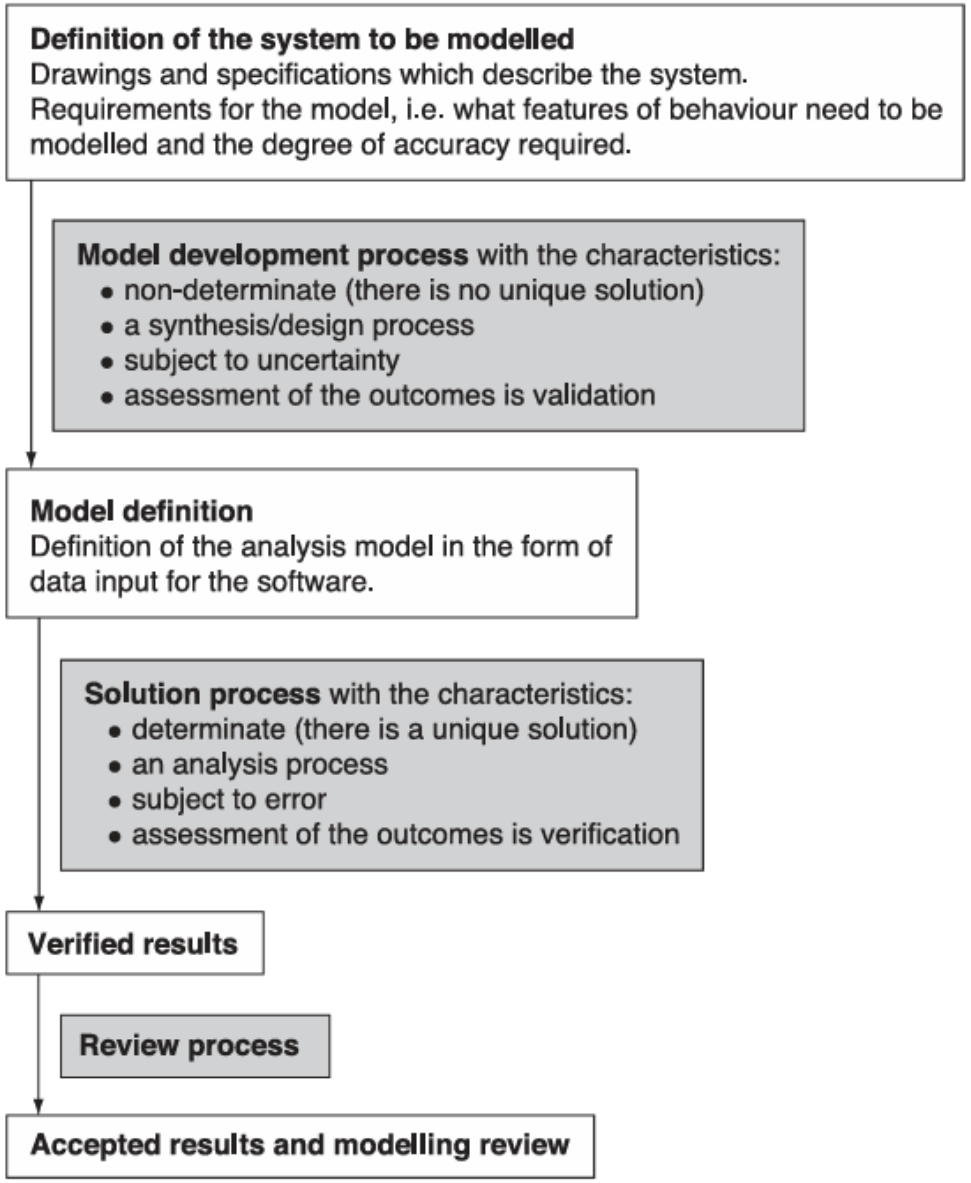
Modern Structural Analysis - Modelling Process and Guidance

MacLeod, Iain A. (Iain Alasdair)

London: ICE Publishing, 2005

❝ This book is essential reading for 21st century practitioners and students who need to do structural analysis...❞

View Online



[1] MacLeod, I. A. (2005)

Verification process

3.6 Verifying the results

3.6.1 *Acceptance criteria for results*

The acceptance criteria for results can be based on risk principles, i.e. on consideration of the combination of likelihood of modelling errors and the consequence of a resulting structural failure – see Table 12.8. The amount of resource to be applied in the checking process will depend on the perceived level of risk. For high-risk situations it may be necessary to commission a complete re-analysis, using different personnel to set up the data and a different software system.

3.6.2 *Verification process*

The questions to be asked in a results verification are: do the results correspond to what is expected from the model?, or have any errors been made in implementing the solution process?

Whereas validation may be less necessary for standard situations, results verification can never be omitted from the process. There is always significant potential for data errors.

[1] MacLeod, I. A. (2005)

Verification process

Checklist

In the results verification process the following checks may be carried out.

- *Error warnings* – check for error warnings in the output.
- *Data check*.
- *Overall equilibrium check* – if the sum of the applied loads on a structure in a given direction is P_a and the sum of the support reactions in that direction is P_s then

$$P_a - P_s = R \tag{3.1}$$

- R should be 0.0, but will normally be a small number, typically of the order of 10^{-12} to 10^{-13}. R is a *residual* of the solution. Reasons for it not being close to zero include:
  - there is ill-conditioning
  - if P_a is calculated using the loading that is expected to be applied (rather than that actually applied as obtained from the software output) and R is not small, then it is likely that there is an error in the loading data.
- *Restraints* – check that the support restraints have been correctly applied by looking at the deformations at the restrained nodes.
- *Symmetry* – if the structure is symmetric then it is worthwhile to apply a symmetric loading case and check the corresponding deformations (or force actions) in relation to a pair of symmetric degrees of freedom. The difference between them should be close to zero, in a similar way to the overall equilibrium check (i.e. use 12-figure accuracy if it is available). If the difference is not small (typically in the last three of the 12 significant digits) then there could be ill-conditioning, but the most likely possibility is that there is an error in the data. For the check to be effective, the symmetry in the definition of the model needs to be precise.
- *Qualitative check* – carry out a qualitative analysis of the results by looking at the deflected shape and the distribution of element forces and stresses. Do these conform with what is expected?
- *Quantitative check* – create a checking model (Section 3.6.3).

[1] MacLeod, I. A. (2005)

Verification process

3.6.3 *Checking models*

A checking model is a simplified, more approximate, version of the main model, but which has adequate accuracy for checking purposes (MacLeod 1990). The results from the checking model should show a correlation with the main model results that is consistent with the degree of approximation made for the checking model. The checking model should be used, if practicable, to check both deformations and element forces and stresses.

[1] MacLeod, I. A. (2005)

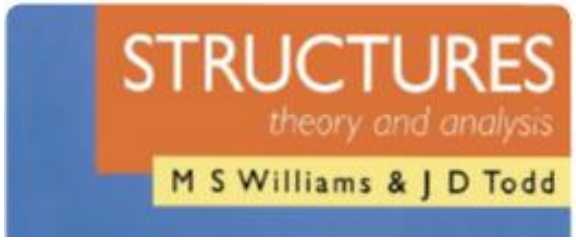
Verification process

The checking model can be a simplified model of the system suitable for 'back of an envelope' calculations or it could require a computer solution. If the checking model is a simplified version of the main model, the results may not correspond closely. It is important to be able to justify the degree of difference between the results from the two models. If this cannot be done, then the checking model may have limited value.

Checking models for non-linear analysis can be difficult to formulate. Running a preliminary elastic analysis and using a checking model might be useful for checking the model in general.

[1] MacLeod, I. A. (2005)

Verification

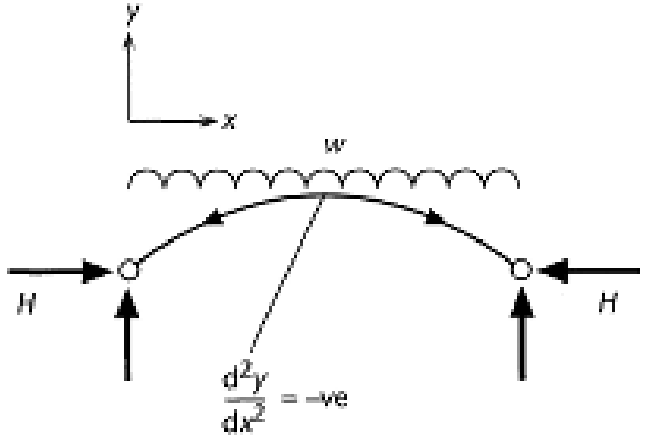


3.4.2 Parabolic arches

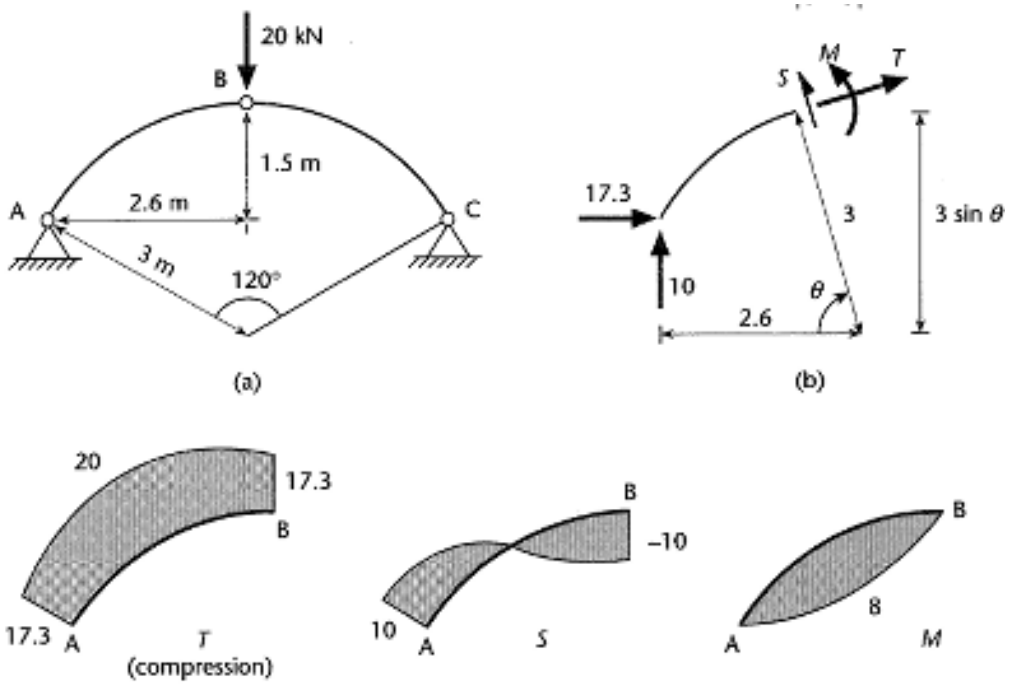
We have shown that a uniformly loaded cable, whose internal force is pure tension, *must* have a parabolic shape. As we saw in section 1.3.4, by simply turning the structure upside down (Figure 3.25), we obtain a parabolic arch, which acts in pure compression. Of course, an arch is not just an inverted cable. Normally it will

3.4.3 Three-pinned arches

Parabolic arches, carrying loads in pure compression, are common. However, other arch shapes are also quite frequently used. While these still carry a large proportion of their load in compression, they also sustain bending moments and shear forces. Non-parabolic arches with pinned ends are redundant, since they possess four reactions, for which only three equations can be formulated. But if we introduce a third pin within the arch then an equation of condition is available, allowing the structure to be analysed. A three-pinned arch is therefore statically determinate.



Arch

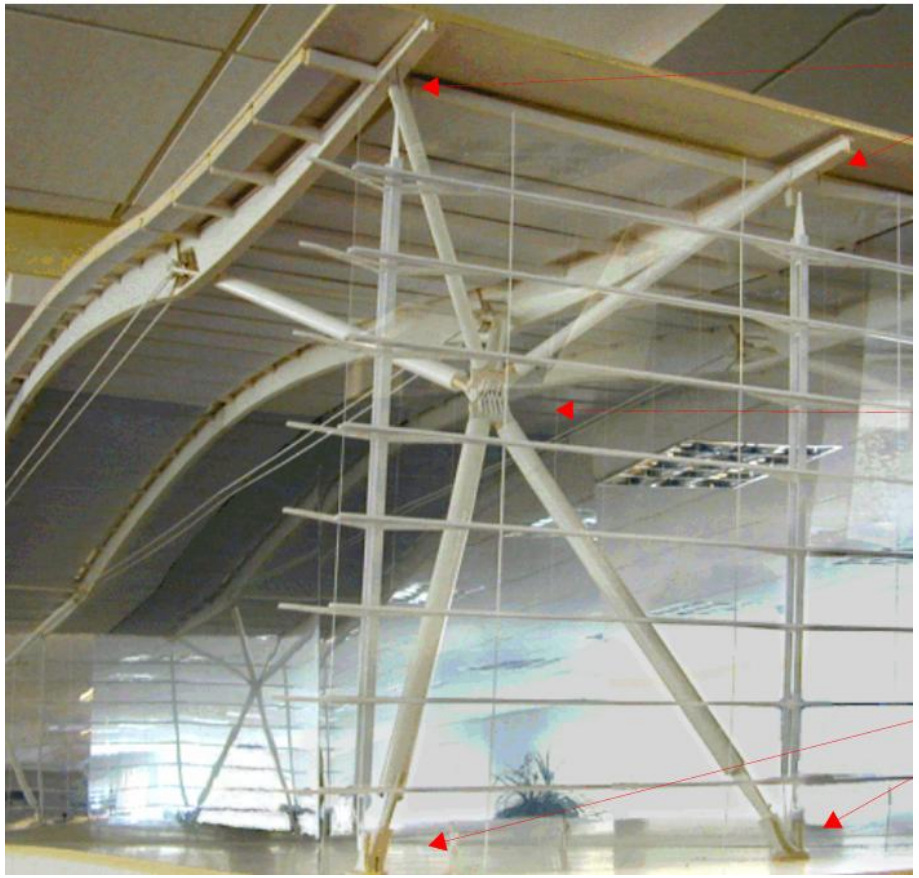


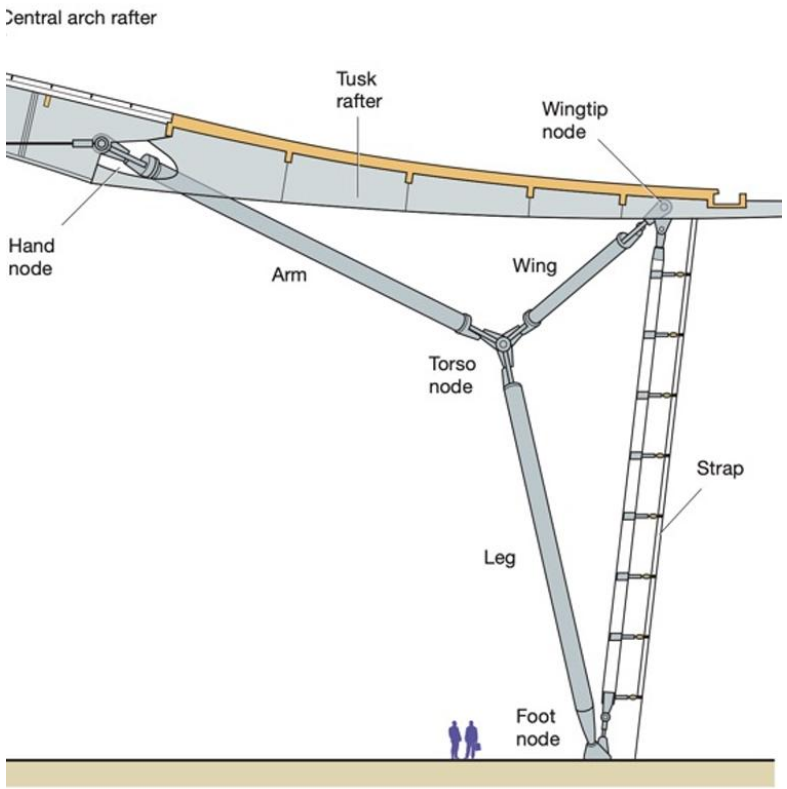
[2] Williams and Todd, 2020

Notes on the assignment

Modelling simplification

- Going from 3D to 2D





How does this simplification affect:

- Stiffness?
- Force?
- Deflection?

[3] Supplementary material

Construction sequence

5. Influence of Prestressing Assumptions (*1 x A4 page maximum*)

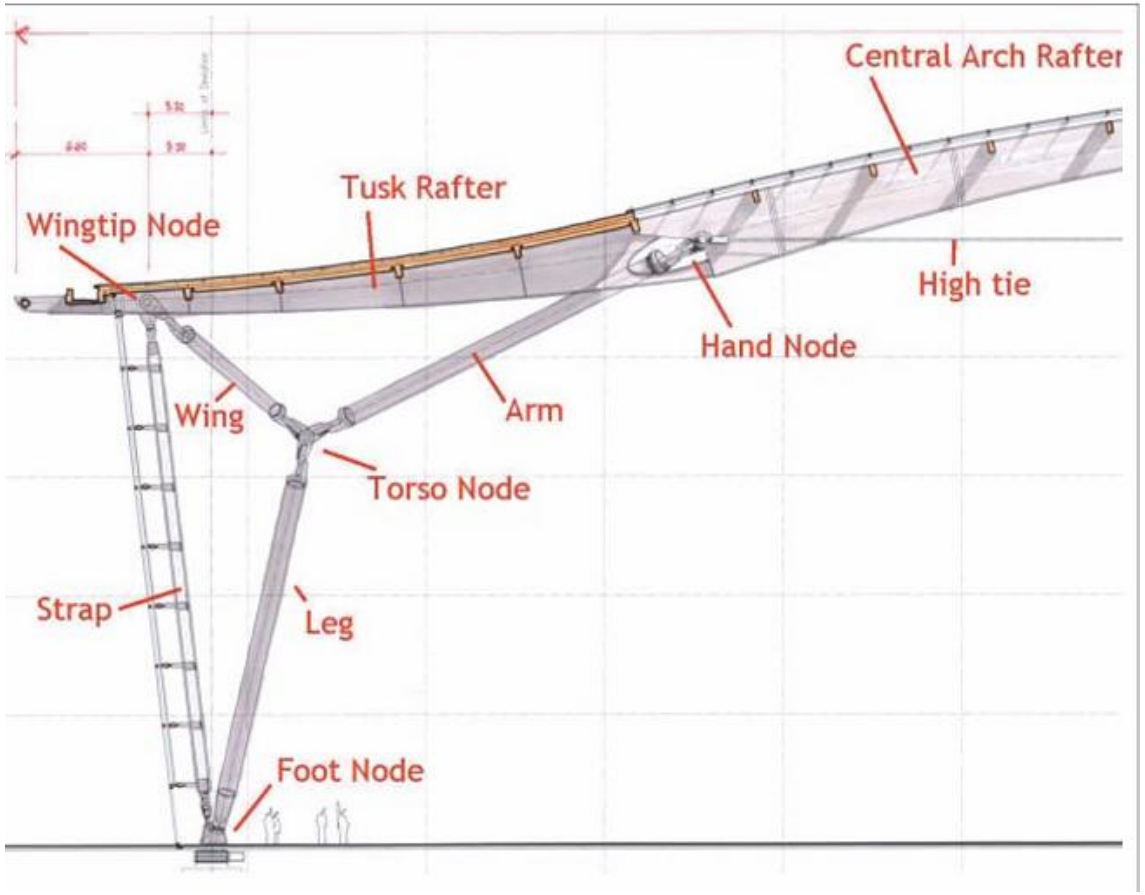
Qualitatively and quantitatively assess the influence of assuming that prestressing of the high tie occurs after full erection, compared with the construction process described in the supplementary documentation.

You should:

- Compare the structural response under the two assumptions
- Comment on differences in internal forces and overall behaviour
- Discuss the implications for design and construction tolerances

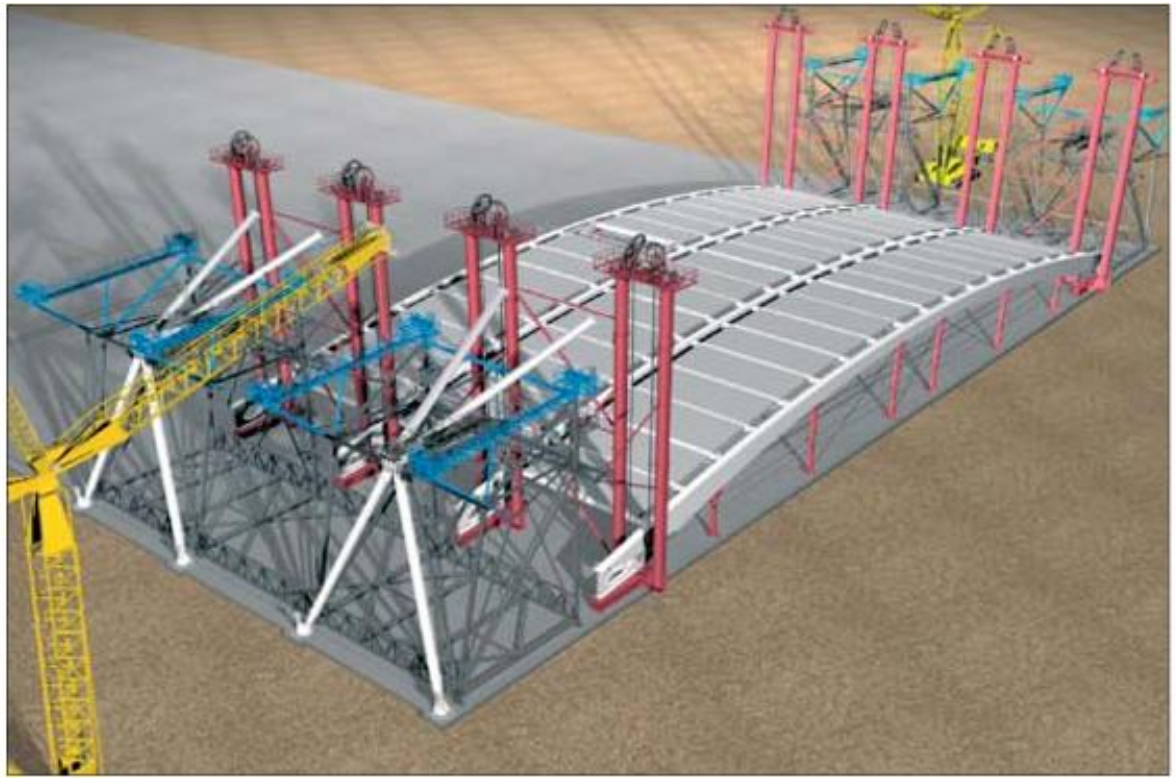
Full construction stage modelling is not required. Comparative reasoning based on simplified models is sufficient.

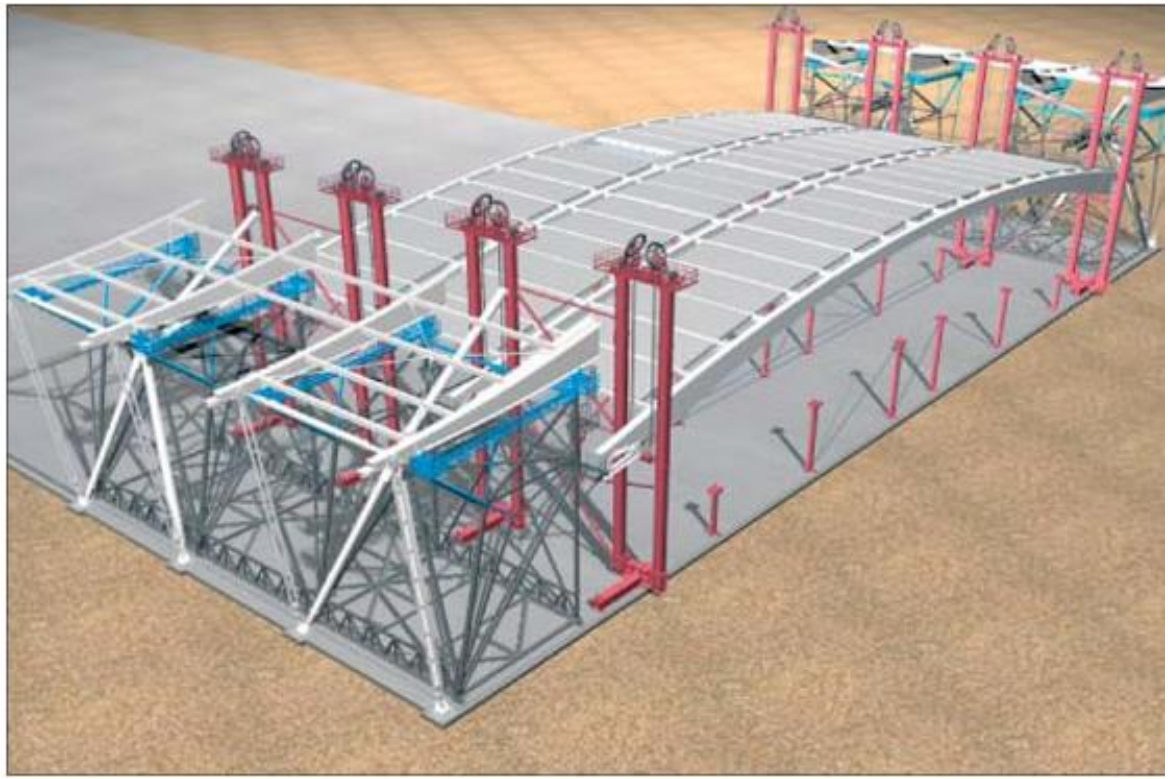
Assumed loading for model



[3] Supplementary material

Actual construction sequence



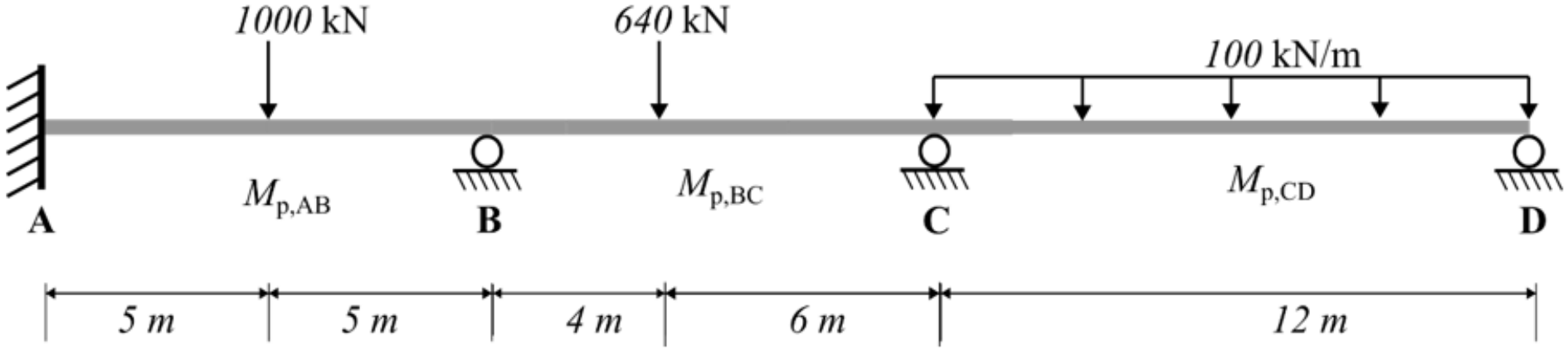


[3] Supplementary material

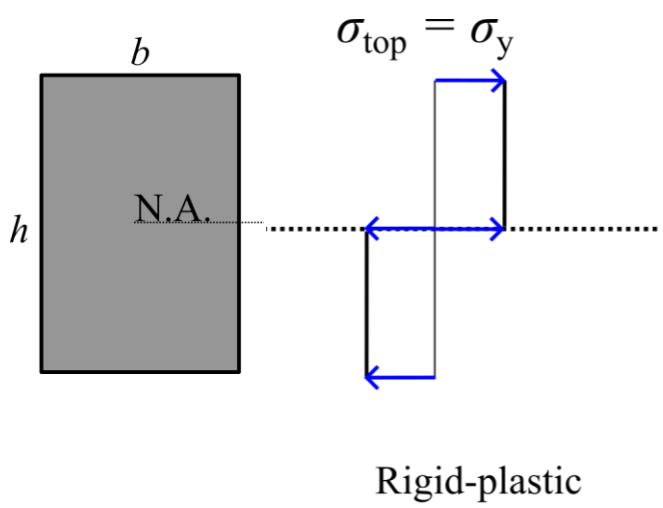
Calculating plastic moment of resistance.

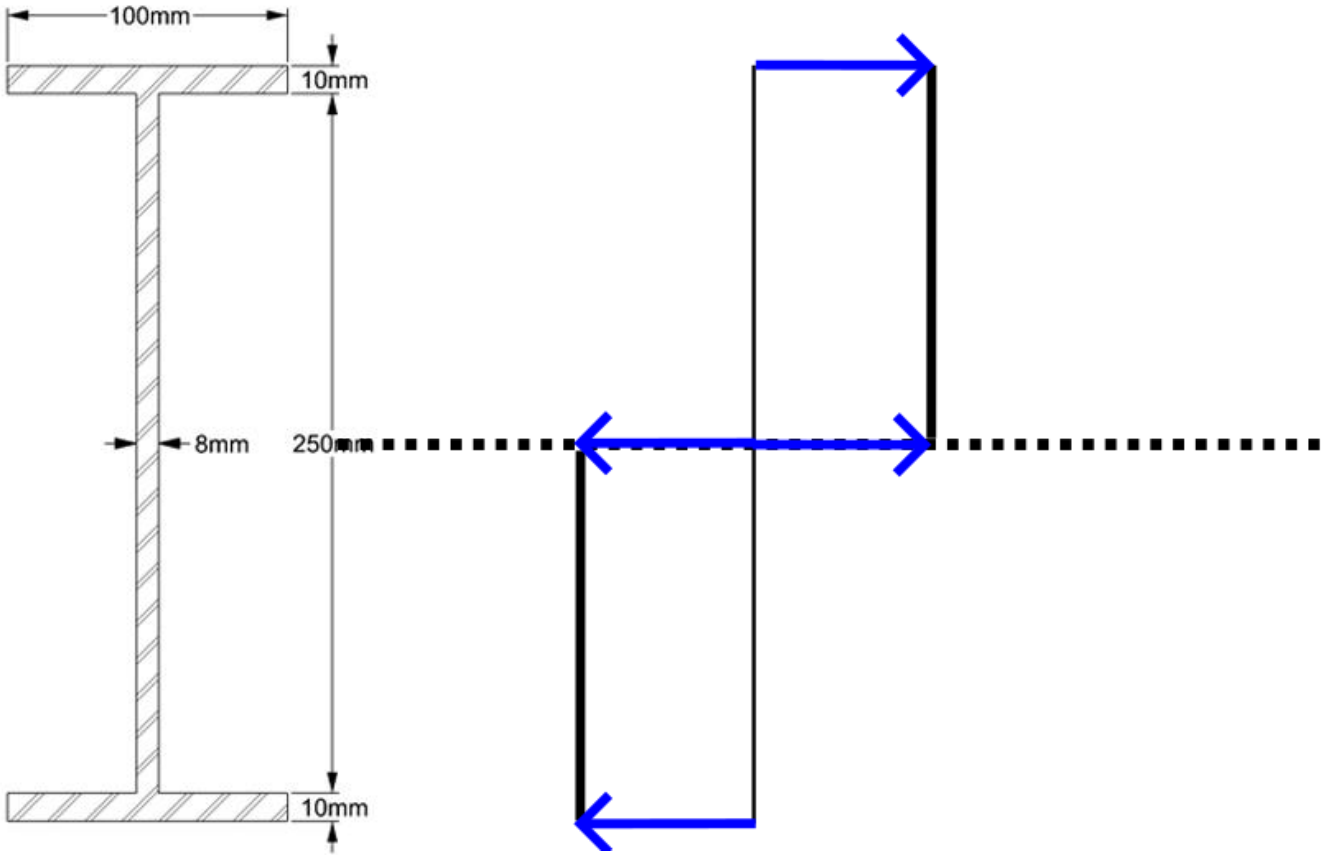
Design requirements

You are required to design the beam such that the solution is economical and practical to construct. **Standard steel universal beam (UB) sections** may be used **with any strengthening methods** using mild steel.



Calculating plastic section capacity





References

[1] MacLeod, I. A. (2005). *Modern structural analysis : modelling process and guidance / Iain A. MacLeod.* Thomas Telford

[2] Williams, M., and J. D. Todd. *Structures: Theory and Analysis*. Macmillan Education UK, 2000.

Research project participants

Full Invitation to Participate Letter

45-60 MINUTE TEAMS INTERVIEW - £20 VOUCHER

1

Participants required for doctoral research. £20 voucher for all those selected to take part.

2

The aim of this study is to investigate undergraduates views and opinions of blended learning in comparisson to those of Academics, Lecturers, or teaching staff across two UK Higher education institutions.

3

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